

Name: Janhvi Sawarkar

PRN: 202501040125

Batch: CS2(C21)

CODETANTRA LAB PRACTICAL 2

The screenshot shows the CODETANTRA lab interface for the 'List operations' task. The task description on the left states: 'Write a Python program that implements a menu-driven interface for managing a list of integers. The program should have the following menu options: 1. Add, 2. Remove, 3. Display, 4. Quit. The program should repeatedly prompt the user to enter a choice from the menu. Depending on the choice selected, the program should perform the following actions: Add: Prompts the user to enter an integer and add it to the integer list. If the input is not a valid integer, display "Invalid input". Remove: Prompts the user to enter an integer to remove from the list. If the integer is found in the list, remove it; otherwise, display "Element not found". If the list is empty, display "List is empty". Display: Displays the current list of integers. If the list is empty, display "List is empty". Quit: Exits the program. The program should handle invalid menu choices by displaying "Invalid choice". Ensure that the program continues to prompt the user until they choose to quit (option 4).' Below the description is a 'Sample Test Cases' input field.

The code editor on the right shows the following Python code:

```
1 list=[]
2 while True:
3     print("1. Add")
4     print("2. Remove")
5     print("3. Display")
6     print("4. Quit")
7     ch=int(input("Enter choice: "))
8     if ch==1:
9         num=int(input("Integer: "))
10        list.append(num)
11        print("List after adding:",list)
12    elif ch==3:
13        if not list:
14            print("List is empty")
15    else:
```

The test results show '2 out of 2 shown test case(s) passed' and '1 out of 1 hidden test case(s) passed'. The test case details show the expected and actual outputs for the menu options and user inputs.

The screenshot shows the CODETANTRA lab interface for the 'Dictionary Operations' task. The task description on the left states: 'Write a Python program to perform insertion, update, deletion, and traversal operations on a dictionary. An initial dictionary containing 10 predefined records is already given in the program. Operations to be Performed: 1. Insertion – Insert a new key-value pair into the dictionary using user input. 2. Update – Update the value of an existing key using user input. 3. Deletion – Delete a specified key from the dictionary using user input. 4. Traversal – Traverse the final dictionary and display all key-value pairs. Input and Output Format: 1. Read an integer representing the key to be inserted and a string representing its value. Insert this new key-value pair into the dictionary and display the dictionary after insertion. 2. Read an integer representing the key to be updated and a string representing the new value. Update the value of the specified key (only if it exists) and display the dictionary after the update. 3. Read an integer representing the key to be deleted. Delete the specified key from the dictionary (only if it exists) and display the dictionary after deletion. 4. Finally, traverse the dictionary and display all key-value pairs. The program should also display the original dictionary before performing any operations. Each output should be printed with appropriate messages. Note: All operations must be performed using dictionary methods. Perform deletion only if possible; leave the dictionary unchanged. Refer to the visible test cases for better understanding and strictly match with the input/outputs.' Below the description is a 'Sample Test Cases' input field.

The code editor on the right shows the following Python code:

```
1 # Initial dictionary with 10 predefined records
2 student = {
3     1: "Amit",
4     2: "Riya",
5     3: "Kiran",
6     4: "Neha",
7     5: "Arjun",
8     6: "Pooja",
9     7: "Rahul",
10    8: "Sneha",
11    9: "Vikram",
12    10: "Anjali"
13 }
14
15 # write your code here...
```

The test results show '2 out of 2 shown test case(s) passed' and '2 out of 2 hidden test case(s) passed'. The test case details show the expected and actual outputs for the dictionary operations and user inputs.

CODETANTRA

HomeLearn Anywhere

202501040125@mitaoe.ac.inSupportLogout

2.2.1. Linear search Technique

Write a program to check whether the given element is present or not in the array of elements using linear search.

Input format:

- The first line of input contains the array of integers which are separated by space
- The last line of input contains the key element to be searched

Output format:

- If the element is found, print the index. If the element found multiple times, print the starting index
- If the element is not found, print **Not found**.

Sample Test Case:

Input:
1 2 3 4 3 5 6
3
Output:
2

Sample Test Cases

CTP1709...

```
1 arr=list(map(int,input().split()))
2 key=int(input())
3 if key in arr:
4     print(arr.index(key))
5 else:
6     print("Not found")
```

Average time0.015 s14.75 msMaximum time0.019 s19.00 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 111 ms

Expected output1 2 3 4 3 5 632

Actual output1 2 3 4 3 5 632

Test case 219 ms

TerminalTest cases

< PrevResetSubmitNext >

CODETANTRA

HomeLearn Anywhere

202501040125@mitaoe.ac.inSupportLogout

2.2.2. Captain of the Team

You are provided with the heights of 11 cricket players (in centimeters). Your task is to identify the tallest player, who will be selected as the captain of the team.

Input Format:

The first line of input will contain 11 integers, each representing the height of a player (in centimeters), each separated by a space.

Output Format

The output should be the height (in centimeters) of the tallest player.

Sample Test Cases

captainof...

```
1 l=input().split()
2 print(max(l))
```

Average time0.006 s6.33 msMaximum time0.009 s9.00 ms

1 out of 1 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 19 ms

Expected output171 169 185 156 174 191 186 190 187 172 160191

Actual output171 169 185 156 174 191 186 190 187 172 160191

TerminalTest cases

< PrevResetSubmitNext >